

Two-Stage Least Squares (TSLS)

ECMT 676 (Econometric II) Lecture Notes

TAMU

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A simple model

- Textbook: Wooldridge, Econometrics of Cross Section and Panel Data, 2nd edition, 2010, MIT press.
- Simple linear model in the population:

$$y = \beta_0 + \beta_1 x + u \quad ((1))$$

where u is thought to be correlated with x (which can have any kind of features – discrete, continuous, or hybrid).

- If $Cov(x, u) \neq 0$ (x is “endogenous”) then ordinary least squares (OLS) will be inconsistent for β_1 .
- If we have a rich set of controls, we might be able to break the link between u and x – but that is an OLS solution.

- An instrumental variable, z , for x has two properties:

$$\text{Cov}(z, u) = 0 \quad (\text{exogeneity}) \quad ((2))$$

$$\text{Cov}(z, x) \neq 0 \quad (\text{relevance}) \quad ((3))$$

- A key difference is that while we must take (2) on faith (or have other ways of checking it), we can always test the null that z and x are uncorrelated given a sample of data (and hope to strongly reject the null that they are uncorrelated).

- Examples of instruments
- Quarter of birth (y: salary, x: education)
- Regional alcohol prices (y: students exam score, x: alcohol consumption, then data should have variation in regions)
- College proximity (x: education)
- Price of shaving cream (y: men's health status, x: whether he shaves everyday)

- Apply the covariance operator to $y = \beta_0 + \beta_1 x + u$:

$$\text{Cov}(z, y) = \beta_1 \text{Cov}(z, x) + \text{Cov}(z, u)$$

and then impose (2) and (3) to get

$$\beta_1 = \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)}. \quad ((4))$$

- Equation (4) establishes that β_1 is *identified* because it is a function of population moments of variables we can observe.

- Replacing the population covariances with the sample covariances: the instrumental variables (IV) estimator

$$\begin{aligned}\hat{\beta}_{1,IV} &= \frac{N^{-1} \sum_{i=1}^N (z_i - \bar{z})(y_i - \bar{y})}{N^{-1} \sum_{i=1}^N (z_i - \bar{z})(x_i - \bar{x})} \\ &= \beta_1 + \frac{N^{-1} \sum_{i=1}^N (z_i - \bar{z})(u_i - \bar{u})}{N^{-1} \sum_{i=1}^N (z_i - \bar{z})(x_i - \bar{x})},\end{aligned}$$

where the second line uses $y_i = \beta_0 + \beta_1 x_i + u_i$, so $y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (u_i - \bar{u})$.

- Consistency of $\hat{\beta}_{1,IV}$ follows by the law of large numbers (for iid data).

- The IV estimator is not unbiased when it is actually needed, that is, when x is correlated with u .
- If $E(u|x, z) = 0$ then, by random sampling,

$$E(u_i|\mathbf{X}, \mathbf{Z}) = 0, \text{ all } i.$$

- It can then be shown that $E(\hat{\beta}_{1,IV}|\mathbf{X}, \mathbf{Z}) = \beta_1$.
- It means the IV estimator is conditionally unbiased, which implies unbiasedness.

If the IV is problematic...

- What happens if the correlation between z and x is small?
- Allowing u to be correlated with both x and z and using, for example, that $Cov(x, u) = Corr(x, u)\sigma_x\sigma_u$, it can be shown that

$$\text{plim } \hat{\beta}_{1,OLS} = \beta_1 + \frac{\sigma_u}{\sigma_x} \cdot Corr(x, u) \quad ((5))$$

and

$$\text{plim } \hat{\beta}_{1,IV} = \beta_1 + \frac{\sigma_u}{\sigma_x} \cdot \frac{Corr(z, u)}{Corr(z, x)} \quad ((6))$$

- If $\text{Corr}(z, x)$ is small – that is, z is a “weak instrument” – then even a small correlation between z and u can produce a larger asymptotic bias than OLS.
- In economics, very common to see IV estimates that are larger in magnitude than OLS estimates.
- And weak instruments lead to large asymptotic standard errors, too.

- Under a homoskedasticity assumption (to be made precise later),

$$Avar\sqrt{N}(\hat{\beta}_{1,IV} - \beta_1) = \frac{\sigma_u^2}{\sigma_x^2 \rho_{z,x}^2}. \quad ((7))$$

- so the asymptotic standard deviation of $\sqrt{N}(\hat{\beta}_{1,IV} - \beta_1)$ is

$$\frac{\sigma_u}{\sigma_x} \cdot \frac{1}{|\rho_{z,x}|}$$

- When $|\rho_{z,x}|$ is small, the asymptotic variance can be very large.
- The formula for the OLS estimator omits $|\rho_{z,x}|$.
- The derivation of (7) can be done in the general multiple regression.
- Weak instruments also make the usual asymptotic approximations highly suspect, even with large sample sizes.

Example

Estimating the return to schooling via simple regression:

$$\log(wage) = \beta_0 + \beta_1 educ + u \quad ((8))$$

Suggestions for z :

- (i) mother's education
- (ii) number of siblings
- (iii) distance to the nearest college at age 16
- (iv) last digit of Social Security Number
- (v) z binary, $z = 1$ if born in first quarter of year
- (vi) z is a randomly assigned education grant during high school

A general treatment of the IV estimator

- Start again with the population model

$$y = \mathbf{x}\boldsymbol{\beta} + u, \quad ((9))$$

where $\mathbf{x}$ is $1 \times K$, $\boldsymbol{\beta}$ is $K \times 1$, and in the vast majority of cases, $x_1 = 1$.

- This is the same *model* that we *estimated* by OLS.
- No such thing as an OLS “model” or IV “model.” OLS and IV are different *estimation* methods that can be applied to the same model. They are consistent under different assumptions.

- Let $\mathbf{z} = (z_1, z_2, \dots, z_L)$ be a $1 \times L$ vector, where $z_1 = 1$ almost always.
- Further, $\mathbf{z}$ contains all exogenous elements of $\mathbf{x}$.
- But, if one or more elements of $\mathbf{x}$ is correlated with u , $\mathbf{z}$ must contain some outside variables.
- $\mathbf{z}$ is exogenous;

$$E(\mathbf{z}'u) = \mathbf{0}. \quad ((10))$$

How does this assumption – these moment or orthogonality conditions – allow us to identify β ?

- Suppose $L = K$.
- For example, $\mathbf{x} = (1, x_2, \dots, x_{K-1}, x_K)$ and $\mathbf{z} = (1, x_2, \dots, x_{K-1}, z_1)$, so that only x_K is (possibly endogenous) and z_1 as an IV for x_K .
- Using (9) and (10),

$$\begin{aligned} E(\mathbf{z}'\mathbf{y}) &= E(\mathbf{z}'\mathbf{x})\boldsymbol{\beta} + E(\mathbf{z}'\mathbf{u}) \\ &= E(\mathbf{z}'\mathbf{x})\boldsymbol{\beta} \text{ by (10).} \end{aligned} \tag{11}$$

- If we assume the *rank condition*

$$\text{rank } E(\mathbf{z}'\mathbf{x}) = K \quad ((12))$$

then

$$\beta = [E(\mathbf{z}'\mathbf{x})]^{-1}E(\mathbf{z}'y), \quad ((13))$$

which extends the moment condition for OLS (which is the special case $\mathbf{z} = \mathbf{x}$).

- Give a random sample,

$$\hat{\beta}_{IV} = \left(N^{-1} \sum_{i=1}^N \mathbf{z}_i' \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{z}_i' y_i \right). \quad ((14))$$

- Using the algebra of plims and the WLLN, $\text{plim}(\hat{\beta}_{IV}) = \beta$ under (10) and (12).
- IV estimator is just a one-stage estimator (in contrast to a two-stage estimator)

- In matrix notation we can write

$$\hat{\beta}_{IV} = (\mathbf{Z}'\mathbf{X}/N)^{-1}(\mathbf{Z}'\mathbf{Y}/N) = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{Y},$$

where

$$\mathbf{X}_{n \times k} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{pmatrix}, \quad \mathbf{Y}_{n \times 1} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{Z}_{n \times k} = \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \vdots \\ \mathbf{z}_n \end{pmatrix}.$$

- The condition $E(\mathbf{z}'u) = \mathbf{0}$ is called *instrument exogeneity*; the rank condition (12) is also called *instrument relevance*.
- Without extra information we cannot test $E(\mathbf{z}'u) = \mathbf{0}$ because u is unobserved.

Model diagnostics

- We can estimate $E(\mathbf{z}'\mathbf{x})$ so we can test the rank condition.
- Difficult in general, but easy with a single endogenous explanatory variable, x_K . Write the *reduced form* of x_K as

$$x_K = \delta_1 + \delta_2 x_2 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + r_k \quad ((16))$$

where, by *definition*,

$$E(r_K) = 0, \text{Cov}((x_2, \dots, x_{K-1}, z_1), r_K) = 0. \quad ((17))$$

- In other words, the linear projection of x_K on $(1, x_2, \dots, x_{K-1}, z_1)$ is

$$L(x_K | 1, x_2, \dots, x_{K-1}, z_1) = \delta_1 + \delta_2 x_2 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1. \quad ((18))$$

- Can show: the rank condition (12) holds if and only if

$$\theta_1 \neq 0. \quad ((19))$$

- OLS consistently estimates the parameters of a linear projection (not necessarily unbiased).
- Need to reject

$$H_0 : \theta_1 = 0 \quad ((20))$$

in favor of (19) convincingly. Heteroskedasticity-robust inference can be used.

- Do not care about the δ_j in (18), but $x_2, \dots, x_{K-1}$ must be partialled out.
- z_1 could be correlated with x_K , but this is not enough - we require that x_K is *partially* correlated with z_1 .
- The IV estimator is consistent under $E(\mathbf{z}'u) = \mathbf{0}$ if

$$L(x_K | 1, x_2, \dots, x_{K-1}, z_1) \neq L(x_K | 1, x_2, \dots, x_{K-1}).$$

- This is just another way to say that, in a linear sense, z_1 helps to predict x_K controlling for the other exogenous variables.

- Regressing x_K on $1, x_2, \dots, x_{K-1}, z_1$ using the data is often called the **first-stage regression**.
- It should be done to establish sufficient partial correlation between x_K and z_1 .
- Structural form vs. reduced form
- A reduced form also exists for y , and can be written

$$y = \pi_1 + \pi_2 x_2 + \dots + \pi_{K-1} x_{K-1} + \pi_K z_1 + v$$

The π_j can be consistently estimated by OLS.

- If x_K is participation in a program and z_1 is eligibility, the reduced form for y estimates the effect of eligibility on y .
- The “structural” equation, $y = \beta_1 + \beta_2 x_2 + \dots + \beta_K x_K + u$, estimated by IV, attempts to get at the causal effect of the program itself (which is often of central interest).

More than one IVs?

- In some cases, we have more instruments than we need.
- For example, if we can use mother's education as an IV, why not father's education, too?
- Again write

$$y = \mathbf{x}\beta + u \quad ((21))$$

$$E(\mathbf{z}'u) = \mathbf{0} \quad ((22))$$

where $L = \dim(\mathbf{z}) \geq \dim(\mathbf{x}) = K$.

- When $L > K$, have more than one IV estimator. We say the model (21) is (potentially) **overidentified**.
- When $L = K$ and the rank condition holds, the model is **just identified**.
- Suppose z_1 and z_2 are IVs for x_K . Which should we use?
- Under a homoskedasticity assumption, the best IV for x_K is the linear combination of *all* exogenous variables defined by the linear projection.

- In general, the best vector of IVs for $\mathbf{x}$ is the vector of linear projections of each element of $\mathbf{x}$ on $\mathbf{z}$.
- Write the LPs in error form as

$$\underset{1 \times K}{\mathbf{x}} = \underset{1 \times L}{\mathbf{z}} \cdot \underset{L \times K}{\Pi} + \underset{1 \times K}{\mathbf{r}}$$

where Π is the $L \times K$ matrix

$$\Pi = \underset{L \times L}{[E(\mathbf{z}'\mathbf{z})]^{-1}} \underset{L \times K}{[E(\mathbf{z}'\mathbf{x})]}$$

and

$$E(\mathbf{z}'\mathbf{r}) = \mathbf{0}.$$

- For each x_j we can write

$$x_j = \mathbf{z}\boldsymbol{\pi}_j + r_j \equiv x_j^* + r_j$$

where $\boldsymbol{\pi}_j$ ($L \times 1$) is the j^{th} column of Π .

- For any x_j in $\mathbf{z}$, $x_j^* = x_j$, so exogenous variables act as their own instruments.
- In the general case, use

$$\mathbf{x}^* = \mathbf{z}\Pi$$

as the $1 \times K$ vector of instruments for $\mathbf{x}$.

- Because $\mathbf{z}$ is exogenous, so is $\mathbf{x}^*$:

$$E(\mathbf{x}^{*'}u) = \mathbf{0}.$$

- The rank condition becomes

$$\text{rank } E(\mathbf{x}^{*'}\mathbf{x}) = K.$$

But

$$E(\mathbf{x}^{*'}\mathbf{x}) = \Pi\Pi'E(\mathbf{z}'\mathbf{x}) = E(\mathbf{x}'\mathbf{z})[E(\mathbf{z}'\mathbf{z})]^{-1}E(\mathbf{z}'\mathbf{x}).$$

- Under the regular conditions next page, β is identified:

$$\beta = [E(\mathbf{x}^{*'}\mathbf{x})]^{-1}E(\mathbf{x}^{*'}y),$$

- Formally, here are the first two assumptions for 2SLS, stated in the population. (We assume access to a random sample).

Assumption 2SLS.1 (Exogenous Instruments): $E(\mathbf{z}'u) = \mathbf{0}$.

Assumption 2SLS.2 (Rank Condition): (a) $\text{rank } E(\mathbf{z}'\mathbf{z}) = L$; (b) $\text{rank } E(\mathbf{z}'\mathbf{x}) = K$.

- Part (a) rules out perfect collinearity among the exogenous variables (which means we cannot use linear combinations of exogenous variables as additional instruments).
- Part (b) is the practically important restriction, and requires $L \geq K$.

2SLS

- Deriving 2SLS: With

$$\beta = [E(\mathbf{x}^*'\mathbf{x})]^{-1}E(\mathbf{x}^*y),$$

need to worry about unknown Π because $\mathbf{x}_i^* = \mathbf{z}_i\Pi$.

Two-step estimation (2SIV):

- (1) Run the regression $\mathbf{x}_i$ on $\mathbf{z}_i$, $i = 1, \dots, N$ to obtain $\hat{\Pi} = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}$. Obtain the vector fitted values,

$$\hat{\mathbf{x}}_i = \mathbf{z}_i\hat{\Pi}, \quad i = 1, \dots, N.$$

(This is the same as regressing each element of $\mathbf{x}_i$ not in $\mathbf{z}_i$ on $\mathbf{z}_i$, and obtaining the fitted values. Any element of $\mathbf{x}_i$ in $\mathbf{z}_i$ is used as its own fitted value.)

- (2) Use $\hat{\mathbf{x}}_i$ as the vector of IVs for $\mathbf{x}_i$:

$$\hat{\beta}_{IV} = \left(N^{-1} \sum_{i=1}^N \hat{\mathbf{x}}_i' \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \hat{\mathbf{x}}_i' y_i \right).$$

- We can write this differently (in a way which 2SLS really means). Because

$$\begin{aligned}\mathbf{x}_i &= \mathbf{z}_i + \mathbf{v}_i \\ \sum_{i=1}^N \mathbf{z}_i' \mathbf{v}_i &= \mathbf{0} \text{ (by OLS FOCs).}\end{aligned}$$

- So

$$\sum_{i=1}^N \mathbf{z}_i' \mathbf{x}_i = \sum_{i=1}^N \mathbf{z}_i' \mathbf{z}_i$$

and then the IV estimator can be written as a two stage least squares estimator (2SLS):

$$\hat{\beta}_{2SLS} = \left(N^{-1} \sum_{i=1}^N \mathbf{z}_i' \mathbf{z}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{z}_i' y_i \right).$$

- The first-stage regression is $\mathbf{x}_i$ on $\mathbf{z}_i$ to get the fitted values, $\hat{\mathbf{x}}_i$.
- The second-stage regression is y_i on $\hat{\mathbf{x}}_i$.

- Both steps in 2SLS are OLS
- The first step in 2SIV is OLS, while the second step is IV.

- However, the two-stage least squares algorithm is **not** really the best way to think about the estimator.
- For one, standard errors from second-stage regression are not correct.
- Also, we will see that using the fitted values as IVs is not the same as 2SLS approach for some panel data applications.
- Finally, the two-step approach can lead to abuse for simple nonlinear models.

- The 2SLS residuals are defined as

$$\hat{u}_i = y_i - \mathbf{x}_i \hat{\beta}_{2SLS},$$

where it is $\mathbf{x}_i$, not $\mathbf{z}_i$, multiplying $\hat{\beta}_{2SLS}$.

- An algebraic fact is that the $\hat{u}_i$ are orthogonal to $\mathbf{z}_i$ in the sample. This is the condition that determines $\hat{\beta}_{2SLS}$:

$$\sum_{i=1}^N \mathbf{z}_i' \hat{u}_i = \mathbf{0}$$

Asymptotics for 2SLS

- Using full data matrices and some algebra, we can write

$$\begin{aligned}\hat{\beta}_{2SLS} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= [(\mathbf{X}'\mathbf{Z})(\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{X})]^{-1}(\mathbf{X}'\mathbf{Z})(\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{Y}) \\ &= \beta + [(\mathbf{X}'\mathbf{Z}/N)(\mathbf{Z}'\mathbf{Z}/N)^{-1}(\mathbf{Z}'\mathbf{X}/N)]^{-1}(\mathbf{X}'\mathbf{Z}/N)(\mathbf{Z}'\mathbf{Z}/N)^{-1}(\mathbf{Z}'\mathbf{U}/N)\end{aligned}$$

where the last expression can be used to show consistency by applying the WLLN to each term, along with the rank condition and $E(\mathbf{z}'u) = \mathbf{0}$

Key Result: Under 2SLS.1 and 2SLS.2, $\hat{\beta}_{2SLS}$ on a random sample is consistent for β .

For inference, it is useful to show

$$\sqrt{N}(\hat{\beta}_{2SLS} - \beta) = \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i^{*'} \mathbf{x}_i^* \right)^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{x}_i^{*'} u_i \right) + o_p(1)$$

where the $\mathbf{x}_i^* = \mathbf{z}_i \Pi$ are the linear projections.

- It follows that

$$\sqrt{N}(\hat{\beta}_{2SLS} - \beta) \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$$

$$\mathbf{A} = E(\mathbf{x}_i^{*'} \mathbf{x}_i^*)$$

$$\mathbf{B} = E(u_i^2 \mathbf{x}_i^{*'} \mathbf{x}_i^*).$$

- Assumption 2SLS.3 (Homoskedasticity): u^2 is uncorrelated with all elements of $\mathbf{z}$ as well as z_j^2 and $z_j z_h$. That is,

$$E(u^2 \mathbf{z}' \mathbf{z}) = E(u^2) E(\mathbf{z}' \mathbf{z}) \equiv \sigma^2 E(\mathbf{z}' \mathbf{z}).$$

- Key Result:** Under 2SLS.1, 2SLS.2, and 2SLS.3,

$$\sqrt{N}(\hat{\beta}_{2SLS} - \beta) \xrightarrow{d} \text{Normal}(\mathbf{0}, \sigma^2 \mathbf{A}^{-1})$$

- Consistent (not unbiased) estimators of σ^2 and $\mathbf{A}$:

$$\hat{\sigma}^2 = (N - K)^{-1} \sum_{i=1}^N \hat{u}_i^2 \xrightarrow{p} \sigma^2$$

$$\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i$$

- Under 2SLS.1, 2SLS.2, and 2SLS.3, we can use

$$\widehat{Avar}(\hat{\beta}_{2SLS}) = \hat{\sigma}^2 \hat{\mathbf{A}}^{-1} / N = \hat{\sigma}^2 (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1}$$

- Heteroskedasticity-Robust Inference:

$$\hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-1} / N = \frac{N}{(N-K)} \left(\sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \hat{u}_i^2 \mathbf{x}_i' \mathbf{x}_i \right) \left(\sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1}$$

which is not the same as using the heteroskedasticity-robust inference in the second stage regression y_i on $\mathbf{x}_i$.

- Interestingly, this is the usual heteroskedasticity-robust variance matrix estimator if we run the OLS regression $\hat{u}_i$ on $\mathbf{x}_i$.
- This follows because $\hat{u}_i$ is orthogonal to $\mathbf{x}_i$ in sample, and so all OLS coefficients are zero.

- Efficiency: 2SLS has the smallest asymptotic variance among all IV estimators using linear functions of $\mathbf{z}_i$ as instruments under 2SLS.1, 2SLS.2, and 2SLS.3. (Only meaningful when $L > K$.)

- Potential pitfalls with 2SLS:

(1) A “little” endogeneity of one or more instruments can lead to large inconsistency if the instruments are weak, that is, only slightly partially correlated with the endogenous explanatory variables (EEVs).

(2) The standard errors of 2SLS can be large. Suppose x_K is the only EEV and homoskedasticity holds.

Example

"Children and Their Parents' Lab Supply: Evidence from Exogenous Variation in Family Size"

by Angrist and Evans, American Economic Review, 1998, page 450-.

y :hours of working per week (hours)

x :number of kids (kids)

z :whether the first two kids have the same sex (same sex)